

4. Your friend Atlas, just having finished the rotational dynamics unit, tells you they believe that if you were to throw a ball at the equator, and The Earth and everything else with it disappeared, the ball would go flying off into space at high speeds due because of the speed it got from The Earth's rotation energy. Design a model, estimate, and conclude whether Atlas is correct in their thinking or not.
5. You are about to start climbing a mountain when a spherical rock suddenly rolls passed you down the mountain (without slipping) with a speed of 200 m s^{-1} . If the mountain has an average of a 15° slope, how tall is it?